

FLEX-UF: Supplementary Methods and Controlled Experiments

Anonymous CVPR submission

Paper ID DRAFT

S1. Scope and Reproduction

This supplement records CPU analyses of the frozen evaluation archive, actual-byte evaluation of frozen interim depth models, and the protocol for the ongoing full-schedule depth study. The main manuscript uses the recorded mixed reconstructions of the shared-latent e15 codec. The new control-path and spatial-assignment analyses reuse its stored source-error tables. They produce candidate decisions for later replay, without executing the codec or measuring new inference times. The independent D2/D4/D6 training continues separately; interim measurements are explicitly labelled and do not stand in for a completed full-schedule depth experiment.

The stand-alone paper repository contains the plotted source tables, source hash manifests, editable figure scripts and inspection results under `data/refresh20260927`. Figures can be regenerated on CPU using `scripts/build_figures.py` and `scripts/build_extended_figures.py`. The parent research repository also retains the scripts that derive these summaries from the raw records and checkpoints. Hashes identify the files inspected during this revision; they do not retrospectively establish the exact executable used by a historical run whose code hash was not recorded.

S2. Tracing the Reconstruction Reference

Two measurement paths need to be distinguished. The source-table producer loads the e15 network and a separate released-weight reference. It computes the candidate tile errors $M_{t,k}$ from e15 and the scalar reference error R from the released synthesis network, both on the padded source. The final-reconstruction evaluator instead loads e15 alone, calls its full-frame synthesis path, crops both full and mixed outputs, and measures their difference in source quality. Its raw field named `psnr_release` is consequently a fine-tuned full-depth reference in the inspected code (Fig. S3).

Metric provenance. The legacy field `db_rgb` is also misnamed. The CTC reader upsamples YCbCr 4:2:0 to 4:4:4 and centres the three planes. Synthesis returns this repre-

Table S1. **A field name does not establish the colour space.** QP32 router maps at the nominal 0.1 dB target. Last column applies explicit YCbCr clipping and RGB conversion/clipping. Values are loss in dB relative to the same e15 full-frame output. These two cases are metric checks.

Frame	Archived field	CPU Δ_{444}	RGB
BasketballPass	0.064751	0.064750	0.057855
BQMall	0.077437	0.077436	0.077396

sentation, and the evaluator computes equal-channel tensor MSE directly, before RGB conversion or clipping. Both the padded selection tables and the cropped MSE-ratio loss therefore use YCbCr 4:4:4. We label this quantity Δ_{444} or “444-MSE loss” throughout. The separately recorded 6:1:1 metric is a weighted average of plane PSNRs after chroma downsampling and clipping, not the same MSE ratio.

Two fixed-map CPU replays with strict checkpoint loading reproduce the archived `db_rgb` values within 1.1×10^{-6} dB (Tab. S1). Converting to RGB changes the result. These checks establish the interpretation of the inspected path; they are not a new 265-pair RGB evaluation. Legacy JSON keys and historical annotations are preserved for traceability; `METRICS.md` records their corrected meaning and the replay hashes. The separate independent-depth validation explicitly converts reconstructions to RGB and must not be pooled with this legacy metric.

A CPU comparison of the two state dictionaries finds 255 shared encoder/entropy tensors with no changes. All 143 inherited synthesis tensors that can be compared have changed, with a maximum absolute parameter difference of 0.2481. Thus the reporting path cannot be identified with released synthesis merely because it executes all twelve trunk blocks. After exact key remapping, all 398 inherited warm-start tensors also match the archived official D12 release. Extra adapter/router tensors are outside this comparison. This second check ties the inspected warm start to the released parameters without assuming that its filename proves provenance.

This distinction limits the interpretation of threshold

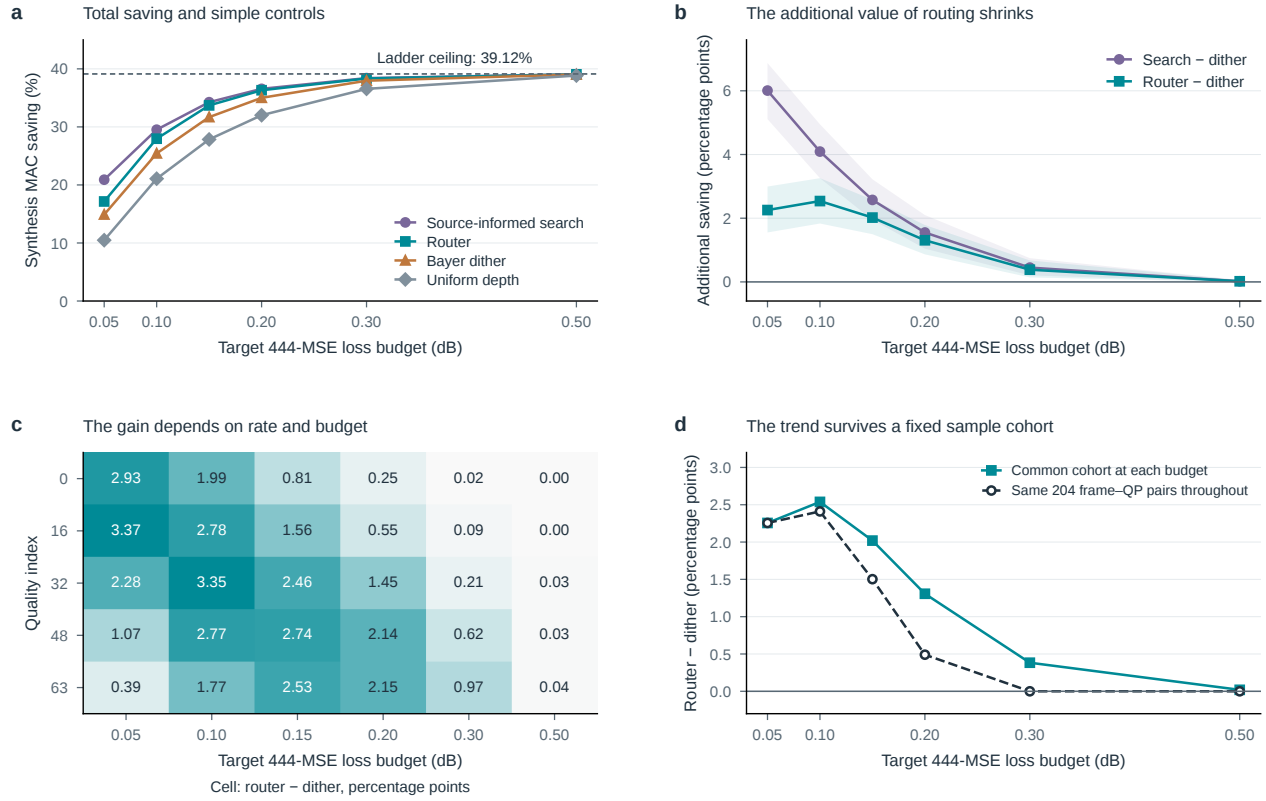


Figure S1. **Budget sensitivity beyond the compact main comparison.** **a,b**, Total saving and paired increments over dithering. **c**, The router increment stratified by QP. **d**, Repeating the comparison on the same 204 frame-QP pairs preserves the contracting margin. Intervals resample sequences and keep QPs together. All policies use per-source calibration; nominal budgets are table targets, not final-image guarantees.

crossings. The recorded 7/263 YCbCr 4:4:4 and 49/263 YUV losses above a nominal 0.1 dB target combine differences in reference, image support, metric and the influence of the shared repair/head. They do not identify the contribution of any single factor. The paired MAC accounting remains attached to the same recorded maps, but the exact quality gap to the cropped released decoder cannot be recovered from the exported JSON alone.

The required replay should retain both full-depth anchors, the padded and valid-pixel errors, and the final mixed reconstruction. Selection and final verification must use the same declared reference and metric. A failed plan should trigger an explicit fallback whose bytes and runtime are recorded.

S3. What Additional Synthesis Depth Buys

Each column of the archived source-error table was produced by decoding a uniform-depth map and measuring per-tile source MSE. Its equal-area tile average is therefore the padded YCbCr 4:4:4 MSE of that actual uniform reconstruction. Comparing this mean to the separately evaluated re-

leased reference gives mean losses of 0.2045, 0.0968, 0.0605 and 0.0379 dB for 6/8/10/12 blocks over all 265 pairs. The corresponding modelled MAC savings are 39.12%, 24.22%, 13.03% and -0.95%. The negative all-deep saving reflects shared repair overhead relative to the full-frame reference. These values use padded support and cannot substitute for cropped released-reference replay.

For adjacent depths, the local quality improvement is $g_t = 10 \log_{10}(M_{t,k}/M_{t,k+1})$. The frame-median improvement, averaged across pairs, is 0.0993 dB for 6-to-8 blocks, 0.0320 dB for 8-to-10, and 0.0213 dB for 10-to-12. The respective mean within-frame interquartile ranges are 0.0748, 0.0340 and 0.0238 dB. This quantifies both diminishing returns and variation within the same image. The average fractions of tiles whose error increases are 4.37%, 11.55% and 11.41%; deeper synthesis is not uniformly better at every location.

Figure S4 uses equal frame-QP weighting, rather than letting larger images dominate through their tile counts. Sequence-cluster intervals retain all five QPs together. Each

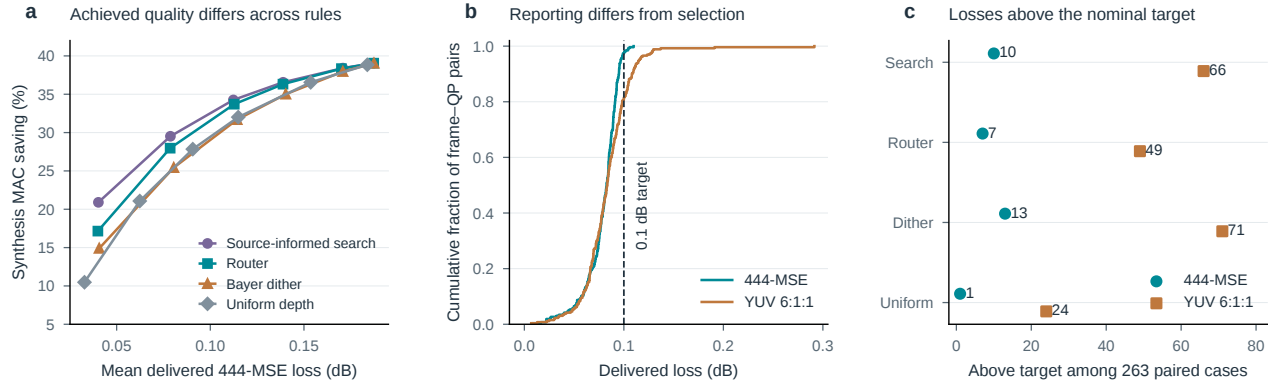
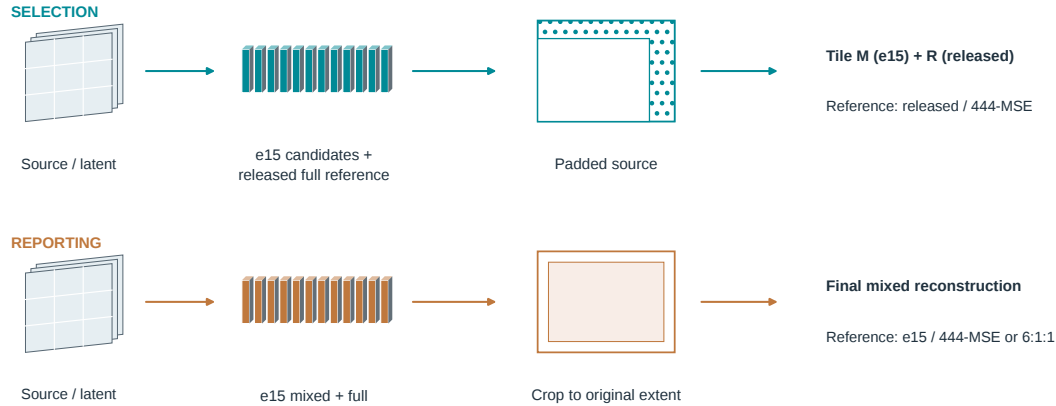


Figure S2. **Nominal targets and final reporting answer different questions.** **a**, Recorded average cropped YCbCr 4:4:4 loss against MAC saving. **b**, Router YCbCr 4:4:4 and YUV 6:1:1 loss distributions at a nominal 0.1 dB target. **c**, Counts above that numerical target, among the 263 common cases and with 10^{-4} dB tolerance. Selection uses a released reference on padded YCbCr 4:4:4; reporting uses cropped e15 full-frame synthesis. The counts therefore do not isolate a common-reference guarantee failure.

a Selection and reporting use different reconstruction anchors and image support



CPU audit: 255 shared encoder/entropy tensors unchanged; 143/143 inherited synthesis tensors changed.

Required comparison: reconstruct both anchors, use the same valid pixels and metric, then verify the selected output.

Figure S3. **Reference and support are part of the measurement.** Candidate tile errors come from e15, whereas the table denominator comes from a separate released-weight reference. Final reporting uses the cropped full-depth e15 reconstruction. The CPU tensor comparison establishes a weight difference in the inspected checkpoints. The figure traces code paths and does not depict newly decoded images.

108 tile comparison still changes the depth of its neighbours be-
109 cause the underlying maps are uniform. It is a conditional
110 reconstruction diagnostic, not a causal one-tile intervention

or a completed independent-codec depth ablation.

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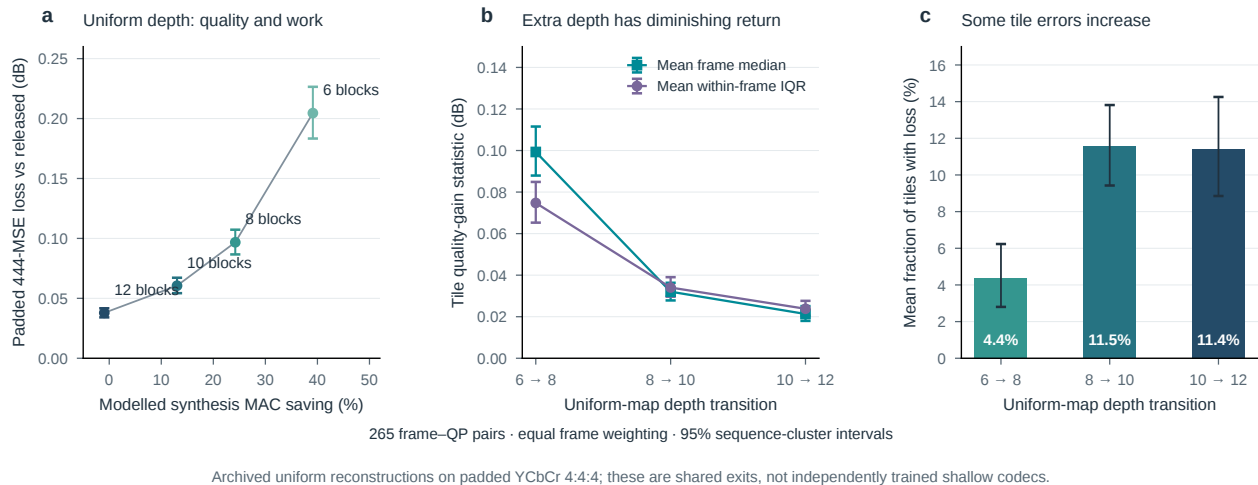


Figure S4. **The value of additional depth varies across regions.** **a**, Uniform-depth padded YCbCr 4:4:4 quality against the released anchor and the recorded MAC model. **b**, Mean frame-median tile improvement and mean within-frame IQR for adjacent depths. **c**, Mean fraction of tiles whose error increases. Error bars are 95% sequence-cluster intervals. These are actual archived uniform reconstructions; neighbouring depths change together, and independent shallow codecs are not represented.

S4. Auditing the Scalar Control Search

For router scores $\ell_{t,k}$ and costs c_k , increasing the price β in $k_t(\beta) = \arg \max_k (\ell_{t,k} - \beta c_k)$ cannot increase selected cost. To see this, write the optimality inequalities at $\beta_1 < \beta_2$ for their respective choices k_1, k_2 and add them. The result is

$$(\beta_2 - \beta_1)(c_{k_1} - c_{k_2}) \geq 0. \quad (\text{S1})$$

No corresponding statement follows for source error $M_{t,k}$, because that quantity is absent from the score being optimised. A feasibility test at the most expensive endpoint can therefore miss a feasible middle assignment, and distortion-based bisection can miss a later feasible interval.

We enumerate pairwise score-line intersections, representatives of the open intervals between them, and the endpoints of the archived range $[-20000, 20000]$. Archived controls are included explicitly to retain their floating-point tie behaviour. Identical maps are merged. Among feasible maps, the selected candidate maximises modelled MAC saving, with table loss as a tie-breaker. This is a finite-path audit in float64, rather than an optimiser over all possible tile assignments.

The cost is monotone on every inspected router path, while source loss has at least one reversal on 227/265 paths. At 0.05 dB, enumeration improves 3 of 204 previously feasible plans, by 0.1626 percentage points on average, and recovers one additional feasible case. At 0.1 dB it improves 2 of 263 plans, by 0.0277 points on average. No paired saving changes at the four larger targets. The largest missed

saving is 30.45 points at 0.05 dB, so a small mean difference should not be read as a uniform per-frame bound.

We strengthen the blind baseline in the same way. For ordered dithering, all changes occur at a finite set of Bayer-rank thresholds and rung boundaries. Enumerating those maps reveals source-loss reversals on 232/265 paths. It improves 3/204 paired plans at 0.05 dB and 1/263 at 0.1 dB, with mean gains of 0.0192 and 0.0028 points, respectively. Six additional cases become feasible at 0.05 dB. The quality of these newly selected maps has not been measured after final mixed reconstruction, so they do not replace the main manuscript’s reconstruction curves.

The audit reproduces every archived feasible control and cost, verifies cost monotonicity, and checks that enumeration never discards a better archived feasible saving. An independent three-exit example verifies the non-monotone failure mechanism against a dense control grid.

S5. Sequence-Disjoint Control Calibration

The source-calibrated sweep answers which allocation a policy can attain when its scalar control is tuned for each image. It does not establish what happens when that control is fixed before seeing a new source image. We therefore add a CPU diagnostic that keeps the router checkpoint and archived tile tables fixed while separating control calibration by sequence. All five QPs of a sequence stay in one fold. Sorting sequence-name hashes and assigning ranks modulo five yields folds of 10/10/11/11/11 sequences. For each QP,

controls are fitted on four folds and evaluated on the fifth; each of the 265 frame-QP pairs is evaluated exactly once per setting. The data have previously informed development, so this is a calibration diagnostic, not an untouched external test.

We search a fixed router grid of 4,097 prices in $[-256, 256]$, which reaches both all-shallow and all-deep endpoints on every frame. Dither uses 385 mixture controls from zero to three; uniform uses the four reachable depths. Among controls satisfying a training-fold mean or 90th-percentile table-loss target, we maximise mean MAC saving. If none satisfies the target, we use the lowest training-risk control and explicitly retain its infeasibility. No test-source-conditioned fallback is applied. A held-out-label perturbation check confirms that test losses cannot change the fitted control.

At a 0.1 dB mean target, router and dither save 27.379% and 24.608% MACs, with mean held-out losses of 0.1006 and 0.0997 dB. Nevertheless, 122/265 and 123/265 cases exceed the target. Fitting the training-fold 90th percentile reduces these counts to 32 and 29, while savings become 17.586% and 15.558%. At the tighter 0.05 dB Q90 target, calibration groups covering 159/265 test cases are infeasible for each policy; these cases remain in every reported aggregate.

This distinction changes the deployment question: an average quality target, a bound on the frequency of large losses, and encoder-side verification of every frame are different contracts. Our table result supports investigating source-free control, but repair, cropping and reference alignment still require final-reconstruction replay before a delivered-quality claim. Section S15 performs this replay at the predeclared QP32; it also reports the unresolved paired routing margin.

S6. Separating the Histogram from Spatial Placement

A map can save arithmetic by selecting many cheap exits even if their locations contain no useful information. To isolate placement, let $p_k = N^{-1} \sum_t \mathbf{1}[k_t = k]$ be its fixed exit histogram and let π be a uniformly random permutation of its assignments. Each tile has marginal exit probability p_k , giving the exact expectation

$$\mathbb{E}_\pi \left[\frac{1}{N} \sum_t M_{t, k_{\pi(t)}} \right] = \frac{1}{N} \sum_t \sum_k p_k M_{t, k}. \quad (\text{S2})$$

We report $G = 10 \log_{10}(\mathbb{E}_\pi[\overline{M}_\pi]/\overline{M})$, where $\overline{M}$ is the recorded map's table MSE. Positive G indicates useful spatial placement at exactly the same histogram and MAC cost. The logarithm is outside the expectation: this is not expected PSNR over random reconstructions.

At 0.1 dB, source-informed search has a mean placement gain of 0.0321 dB, the router 0.0213 dB, and dithering

0.0007 dB over their respective shuffled histograms. The router's 95% sequence-cluster interval is $[0.0167, 0.0262]$ dB; dithering's is $[-0.0002, 0.0017]$ dB. Uniform-depth maps have zero gain by construction. Among the 263 paired router maps, 204 have positive gain, seven have negative gain and 52 use one exit only. This diagnoses content association conditional on a source-calibrated histogram; it does not validate held-out autonomous calibration.

For each map we also solve the exact fixed-histogram assignment problem on M . Repeating each exit column according to its count produces a square linear assignment problem. Its minimum table MSE gives a conditional upper bound on placement value. The router's mean regret to that optimum is 0.0044 dB at the 0.1 dB target. Every source-search map attains its own fixed-histogram optimum, as expected for a separable Lagrangian minimiser. The optimum still requires the complete source table and carries the same information-acquisition cost. Replaying shuffled and optimal maps through the mixed decoder is the next required ablation.

S7. A Common Cap on Delivered YCbCr 4:4:4 Loss

The stored mixed reconstructions permit a second comparison under a common reporting anchor. For each frame-QP pair and policy, we collect its up to six archived nominal-budget maps, deduplicate identical maps, and retain only candidates whose measured cropped YCbCr 4:4:4 loss meets the chosen cap, with tolerance 10^{-4} dB. The evaluated full-frame e15 reconstruction is an explicit fallback with zero relative loss and zero MAC saving. The cheapest eligible candidate is selected; ties prefer lower loss.

This keeps all 265 pairs at every cap. At 0.1 dB, the retained maps save 29.19% of modelled synthesis MACs for source-informed search, 27.75% for the router, 24.81% for dithering and 22.98% for uniform depth. The router-dither difference is 2.93 percentage points, with a paired 95% sequence-cluster interval of $[2.11, 3.79]$. Their mean achieved YCbCr 4:4:4 losses are 0.0786 and 0.0791 dB, respectively. Full-frame fallback is selected for one router case, two search cases, two dither cases and eleven uniform cases. At a 0.3 dB cap, the routing margin falls to 0.35 points $[0.11, 0.64]$ (Fig. S7).

Every selected output was measured in the original archive, and no new image or interpolated operating point is introduced. This diagnostic is nevertheless retrospective and source-aware: it selects among candidates after their final quality is known. It neither accounts for the cost of obtaining those candidates nor establishes an autonomous policy. Its finite candidate pools are not exhaustive, and its reference is fine-tuned e15, not the released model. Within those boundaries, the observed routing margin survives both a common delivered-loss cap and explicit fallbacks.

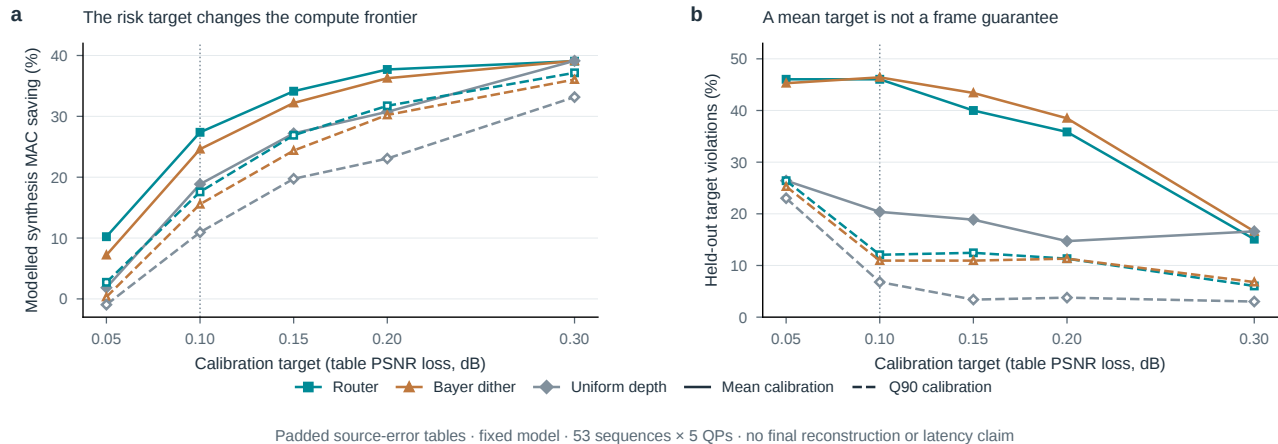


Figure S5. **A source-free control trades arithmetic savings against tail risk.** Controls are calibrated on other sequences, separately at each QP, using the training mean (solid) or 90th percentile (dashed). All 265 pairs are retained, including cases from infeasible calibration groups. **a**, Modelled MAC saving. **b**, Fraction exceeding the table-loss target. These padded source-error diagnostics do not establish final mixed-image quality, a per-frame guarantee or complete decoder speed.

A descriptive influence check removes each sequence with all five QPs. At 0.1 dB, the remaining mean router-dither margin ranges from 2.80 to 3.12 points; this sensitivity range is not a confidence interval. Each QP also has a positive mean margin. Removing the three pairs where either method falls back leaves 2.91 points over 262 pairs, although this restricted subset changes the estimand. At 0.5 dB, 262 pairs tie and all three positive differences belong to one sequence. These checks are reproducible from `cap_influence_audit.json`.

S8. Shared-Exit Training Provenance

The released warm start, shared-exit fine-tuning and independent depth trainings are three distinct checkpoints/protocols. The archived shared-exit metadata lists 379,614 training images, batch size eight and 16 configured fine-tuning epochs. The inspected launcher selects 512-pixel crops through the late part of the upstream schedule, freezes the encoder/entropy path, and enables mixed-tile training, depth-scaled adapters, grid repair, adjacent feature distillation and a full-depth anchor. Its anchor and distillation coefficients are 10 and 1, respectively; the anchor code scales its batch-mean reconstruction error by the batch-mean λ unless the separate per-sample option is enabled. The e15 checkpoint stores weights, epoch and architectural configuration, not the complete launch command. Thus the launcher and metadata document the inspected intended setup, rather than proving every historical process argument.

The evaluated router checkpoint explicitly stores live inputs `stem`, `qp`, soft-label cross-entropy, temperature ratio 0.1, training cost weight 1.3×10^{-5} , 48 stem projection channels and 256 hidden units. It has no spatial-context

convolution. General-purpose latent/scale/bit pathways remain in the class, but do not provide variable content to this checkpoint. The decoder feature is pooled by tile mean and standard deviation, and the QP is divided by 63. Checkpoint inspection is recorded in `design_audit.json`. This corrects the broader description one would obtain by reading the router class defaults alone.

The companion run record matches the checkpoint path, codec epoch, live inputs and stored evaluation statistics. It records 6,000 updates, batch size six, 512-pixel crops, seed zero, initial learning rate 10^{-3} and a regret coefficient of one. The inspected trainer uses AdamW with weight decay 10^{-4} , cosine scheduling and gradient clipping at norm one on Open Images. Its online agreement monitor withholds tiles within training images; the final label-agreement pass draws fresh images from the same shuffled epoch. Neither statistic is used as the manuscript's reconstruction or held-out calibration result. The stored head contains 148,176 parameters, including its inactive input branches.

S9. The Independent Depth Study

The ongoing experiment retains the first two, four or six synthesis trunk blocks of the DCVC-UF intra architecture and trains each complete codec under the pinned official recipe. The encoder and entropy weights learn alongside the decoder. Each model consequently has its own representation, distinct from the shared-latent exits evaluated in the main paper. Figure S9 shows a possible later deployment path, not an implemented six-expert result.

The pinned training protocol. The source is Microsoft's image-training recipe at commit `cbdae87a5445` [4]. All

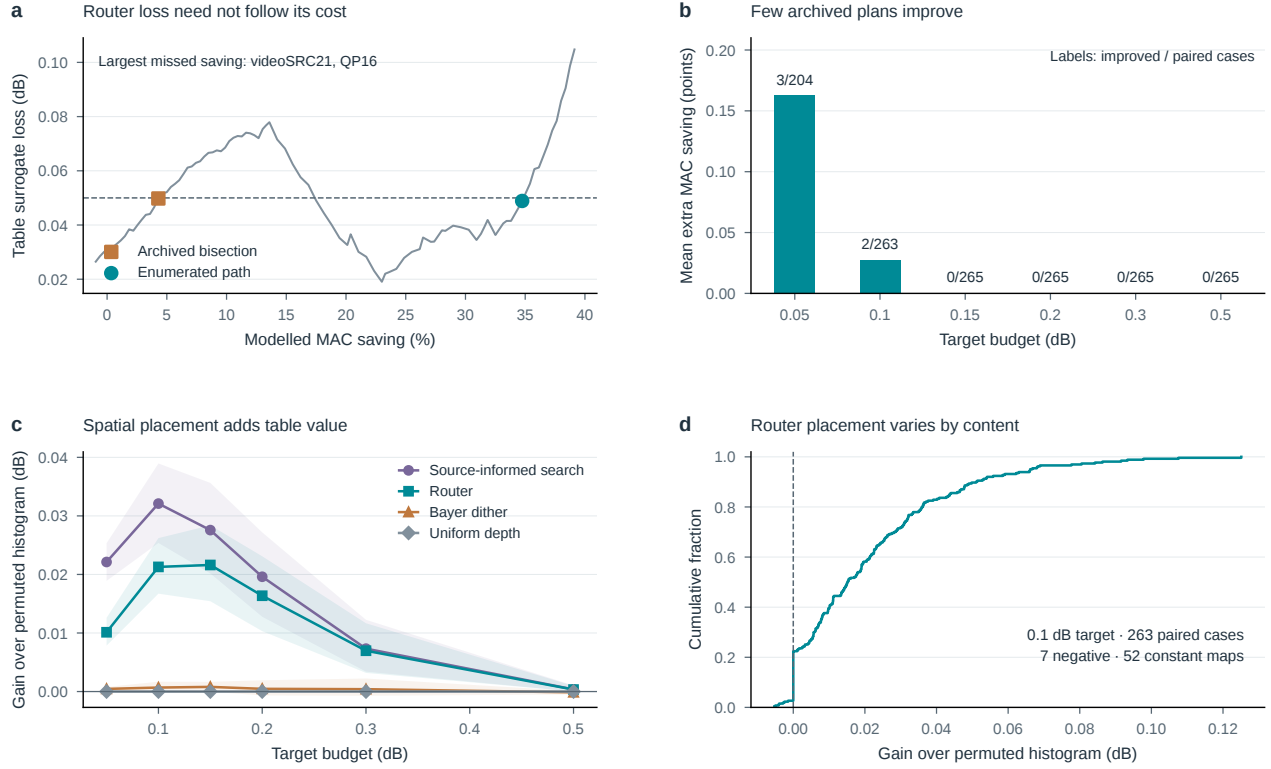


Figure S6. **Control-path and assignment diagnostics.** **a**, The explicitly selected largest missed router saving illustrates non-monotone table loss. **b**, Mean paired improvement from enumerating the router path; labels show improved and paired counts. Newly feasible cases are excluded from these means. **c**, Exact expected-shuffle comparison at unchanged exit histograms, with 95% sequence-cluster intervals. **d**, Router placement-gain distribution at 0.1 dB. All results are CPU analyses of padded error tables; none measures new final-image quality or latency.

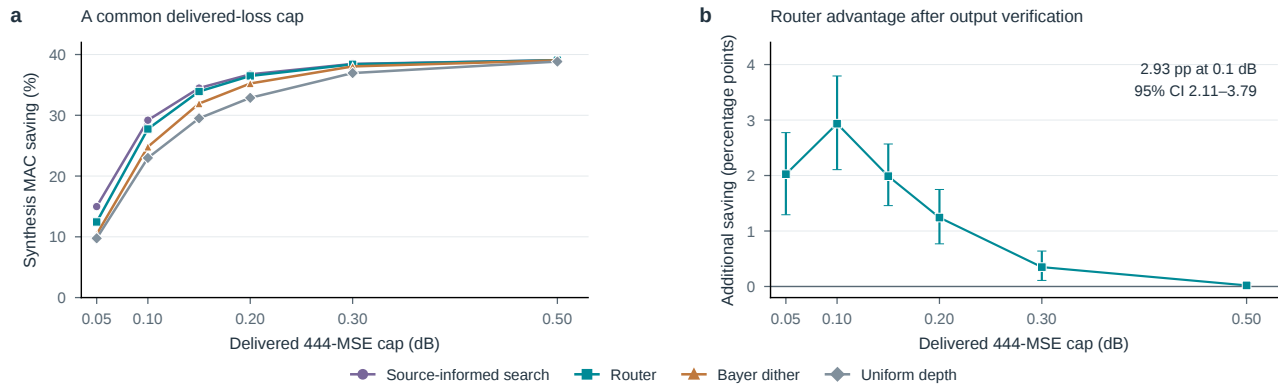


Figure S7. **Controlling delivered loss within the recorded candidate pool.** **a**, Maximum saving among actually measured eligible outputs, including a full-frame e15 fallback. **b**, Paired router–dither difference with 95% sequence-cluster intervals. **c**, Explicit fallback counts. All 265 pairs are retained, with YCbCr 4:4:4-loss tolerance 10^{-4} dB. This retrospective source-aware comparison excludes the cost of candidate generation and makes no released-reference or latency claim.

Table S2. **Unshortened official image-training schedule.** Epoch indices are zero-based. Each row applies to D2, D4 and D6; the architectural intervention is the retained synthesis depth.

Epochs	RGB crop	Learning rate
0–44	256 ²	2×10^{-4}
45–69	256 ²	5×10^{-5}
70–89	256 ²	10^{-5}
90–94	512 ²	2×10^{-4}
95–98	512 ²	5×10^{-5}
99–102	512 ²	10^{-5}
103–104	512 ²	10^{-6}

384,795 images in Open Images train.0/1/2 are retained as lossless decoded-RGB PNGs, with pixel equality checked during preparation. Each independent model uses seed 42, Xavier initialisation, batch size 16, AdamW with the upstream defaults, and gradient clipping at norm 0.1. QP is sampled from 0–63 with logarithmic λ interpolation from 10 to 2048. The original loss, including its literal 0.0833 coefficient, and the upstream crop, flip and colour transforms are retained. Table S2 records the full 105-epoch schedule. There are 24,049 batches per epoch and 2,525,145 scheduled updates per model. With the official shuffled `drop_last=True`, eleven images per epoch do not enter a batch; no image is permanently excluded from the training manifest.

Training uses FP32 with the upstream TF32 flags and `torch.compile`, with `fallback_random=True` to retain the PyTorch random-number path. AMP is disabled. The instrumented eager step was checked against the upstream step at identical inputs and RNG state. Compilation can change floating-point operation order, so protocol parity does not imply bitwise-identical training trajectories. Checkpoint manifests retain the execution mode and source hashes.

Synthesis versus complete neural-decoder arithmetic. We trace every convolution in the pinned architecture, counting each use of the causal spatial prior. At 512×512 , the unchanged neural entropy-recovery path costs 31.079 GMac. D12 synthesis contributes 56.915 GMac, while D2 contributes 14.495 GMac. Thus a 74.53% synthesis reduction becomes a 48.21% reduction when those fixed recovery networks are included (Fig. S8). These are architectural counts, not trained-quality or timing results. Meta-tensor traces were checked against a real CPU D2 forward at 64×64 ; the repeated prior is counted three times. Biases, elementwise operations, rANS, memory movement and routing remain outside this Conv2d accounting.

The native encoder also produces a reconstruction while its CPU entropy worker can run concurrently. Even including synthesis in an encoder MAC total therefore does not justify

adding isolated stage times or converting MAC reduction into wall-time reduction. The same execution contract and precision must be used for every depth in a future paired timing study.

Isolating depth. Released D12 is a useful deployment anchor, but a D12 trained with the same data manifest, 105-epoch schedule, optimisation and initialisation protocol is necessary to isolate depth from training history. D8 and D10 should be prioritised after the initial depth frontier and this control are available. The main runs retain the official recipe; distillation and prefix initialisation are separate variants rather than silent training changes.

Isolating routing. The evaluation should progress from full-frame D12 to patchwise D12, the best validation-selected fixed expert, blind mixtures, simple image statistics, the MLP and source-informed search. Full-frame versus patchwise D12 measures the cost of tiling and entropy resets before routing receives credit for any gain. A histogram-preserving shuffle separates the chosen expert proportions from their spatial placement. A model below the dataset-mean frontier may still help a subset of patches. Expert removal should therefore be evaluated by leave-one-expert-out routing at matched constraints, not justified by mean full-image quality alone.

A concrete router hypothesis. A lightweight encoder-side predictor can estimate each expert’s distortion and rate relative to a cheap reference. Let $z_{t,k}$ indicate the expert assigned to patch t and $n_k = \sum_t z_{t,k}$. For measured grouped-runtime tables $\tau_k(n_k)$, a proposed system objective is

$$\hat{J} = \sum_{t,k} z_{t,k} (\hat{D}_{t,k} + \lambda \hat{R}_{t,k}) + \mu \sum_k \tau_k(n_k). \quad (\text{S3})$$

Routing, packing and scatter/gather work must also enter the measured total; the expression isolates the expert-runtime dependence. An independent per-patch latency label approximates this coupled problem. A bounded post-processing variant could move a patch from k to j according to its predicted RD change and the measured marginal time $\tau_k(n_k - 1) - \tau_k(n_k) + \tau_j(n_j + 1) - \tau_j(n_j)$. The number of passes and decision-time limit should be fixed on validation images. Any improvement must include this extra control work.

Regression or regret-weighted training should be compared against ordinary exit-label cross-entropy. A second-stage top-two verification can be restricted to small predicted margins, with its threshold selected on held-out calibration images and its extra encoding work included. Neither a large prediction margin nor a low average regression error establishes a framework quality guarantee. These are prospective

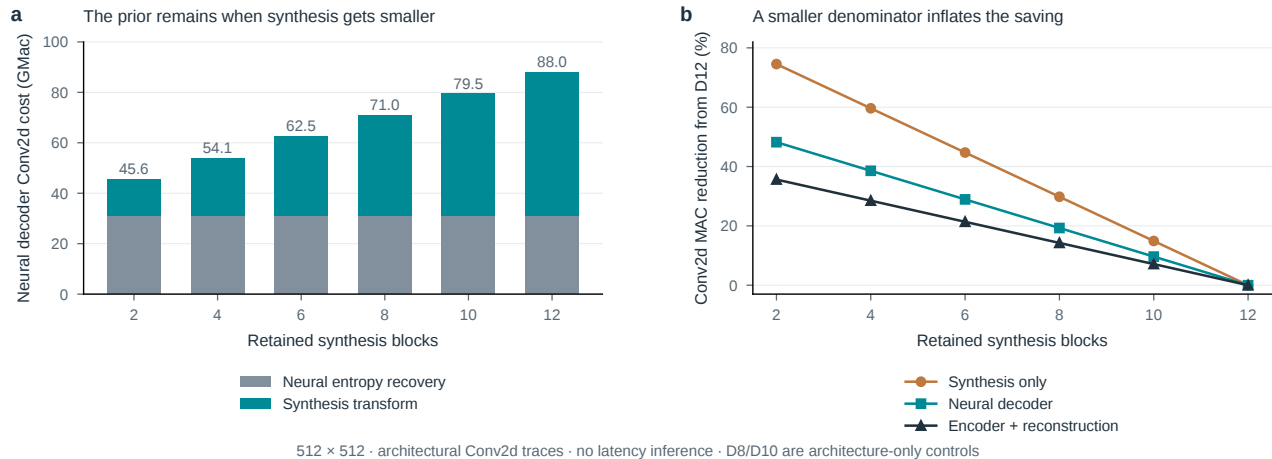


Figure S8. **The arithmetic denominator changes the apparent saving.** **a**, Synthesis varies with depth while neural entropy recovery stays fixed. **b**, The same architectural intervention under three accounting scopes. Counts use 512×512 input and one MAC per multiply-accumulate; they are not observed kernel times. D8/D10 are architecture-only controls. Encoder totals include reconstruction but omit rANS, elementwise work, transfers and control overhead.

experiments, not measured components of the current shared-exit system.

Spatial competition between complete codecs already provides a close precedent [1]; adaptive compression and patch early exit provide further controls [2, 5, 7]. The testable direction here is depth-heterogeneous selection with measured rate-distortion-time costs and reduced encoder search. Practical codec studies motivate including device execution and perceptual assessment [3, 6].

S10. Evaluation Contract for the Next Stage

Expert training, router training, calibration and final evaluation should be split at image level. Patches from the same image must stay in one split. The four fixed validation crops in the current training monitor are health checks; a broader intermediate evaluation should use the available 100 DIV2K validation images, followed by the declared final Kodak/CTC/CLIC scope. The frozen epoch-20 DIV2K crop evaluation below is complete; the final trained-model, full-image and native-runtime study remains pending.

Each codec needs a true-byte rate-distortion curve. Equal QP does not imply equal rate across independently learned codecs. Comparisons should use their common measured support, reporting infeasible targets and avoiding extrapolation. Rate includes main and hyperprior payloads, model identity, quality/control parameters and framing headers, divided by original pixels. Quality is measured after cropping valid pixels; the exact RGB/YUV formula and perceptual metrics are declared before test-set selection.

Timing requires the same GPU, precision, kernel path and warmup, with alternated paired execution order and retained raw trials. Encoder, decoder and complete round-trip times

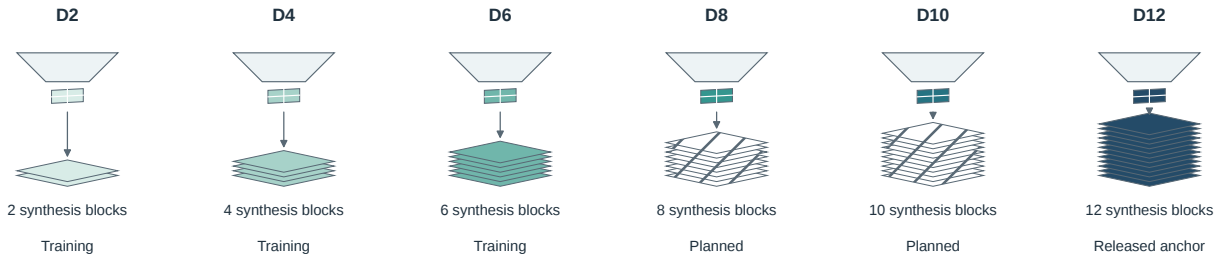
are separate outcomes. CUDA-event module time and end-to-end wall time must remain separately labelled. Grouping, scatter/gather, entropy work, stream parsing, transfers and fallback are included where they occur. The full codec bank's resident weight memory and peak activation memory are both reported.

The primary success criterion is a positive net runtime advantage over the best validation-selected blind control on common achieved rate-distortion support. Critical comparisons should then be repeated across training seeds. A content bootstrap at fixed weights measures a different source of variation and cannot replace those repeats.

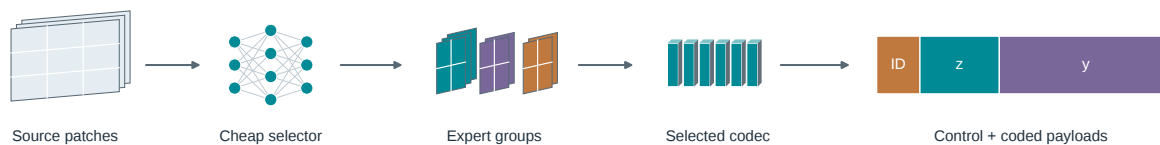
S11. Frozen Interim Depth Models: Actual Coded Rate

The four-crop health monitor cannot establish a rate-distortion frontier. We therefore freeze the D2, D4 and D6 checkpoints after exactly 20 of the 105 scheduled epochs and evaluate all 100 DIV2K validation images as 512×512 centre crops. The released D12 checkpoint is a practical anchor with different training provenance, not the control that isolates the effect of depth. All four models are evaluated at QPs 0, 16, 32, 48 and 63: 2,000 encode/decode cases. No checkpoint is selected using these results. The study is an interim diagnostic, not the final depth ablation.

A causal CPU reference. Each stream contains actual rANS payload. An encoder-independent process reconstructs the image using the stream and resident model, with source analysis disabled. It first decodes the hyperlatent and then the four causal main-latent stages; entropy indexes

a A separate representation at each capacity

All codec weights learn jointly. A matched-training D12 is still required for a depth-only comparison.

b Select before encoding; transmit the selected expert

The decoder uses the signalled expert's entropy model and synthesis; expert latents are not interchangeable.

Charge selection, grouping, entropy resets, assembly and resident model memory to the complete system.

Prospective deployment - glyphs show architecture, not measured routing, throughput or batch support

Figure S9. **An independent codec bank is a separate experimental system.** Training status is shown as of 27 September 2026. D2/D4/D6 are in progress, D8/D10 are planned, and D12 is the released reference. Each expert owns its analysis, entropy and synthesis parameters. The proposed encoder-side MLP selects an expert before encoding; the decoder receives the expert ID. Queues, network shapes and stream segments are schematic. No throughput or quality result is encoded by their geometry.

come from the decoder's recovered context, rather than being supplied by the encoder. Every case has exact reconstruction, latent and symbol/index-trace checks. The decoder process is reused within a model, so independence does not imply a newly launched process for every image. The reference uses CPU FP32 and a versioned research format, not the released CUDA FP16 format or a latency benchmark.

We distinguish entropy estimates, rANS payload and container bytes. The research header is 88 bytes per crop, including 64 bytes of identity and integrity hashes: 0.002686 bpp on a 512^2 crop. This is not a minimal production header or the cost of a routing map. Reconstructions are explicitly converted from clipped YCbCr to clipped RGB before quality measurement. These RGB results are separate from the legacy shared-exit archive's equal-channel YCbCr-MSE metric.

Compare rate, not quality index. For each image, we interpolate PSNR linearly in log payload rate and retain only the common support of all four codecs. No quality is extrapolated. The supported cohorts contain 95, 99 and 89 images at 0.1, 0.2 and 0.4 bpp respectively. At 0.2 bpp, D4 and D6 im-

prove over D2 by 0.1207 dB [0.1018, 0.1420] and 0.1834 dB [0.1572, 0.2115]; the released D12 gap is 0.6986 dB [0.5989, 0.8112]. Intervals use 5,000 paired-image bootstrap draws, conditional on the fixed checkpoints and included images. They do not measure training-seed variation. PCHIP interpolation gives 0.1209, 0.1839 and 0.6880 dB on the same 99-image support. Figure S10 separates native-QP population means from these paired, equal-rate comparisons.

Why the small monitor was insufficient. The original four crops and the additional 96 are reported separately on the same supported cohort (Fig. S11). The four-crop subset overstates the mean depth advantage at 0.2 bpp; it is useful for training health, not a substitute for the validation distribution. D6 nevertheless retains an approximately 0.18 dB mean advantage over D2 across the larger cohort. Its modest magnitude is therefore not solely an artefact of monitoring four images. The remaining released-model gap cannot be attributed to depth until an official-recipe D12 control has been trained for the same schedule.

BD-rate is computed per image using PCHIP interpolation of log rate as a function of PSNR, over the intersection

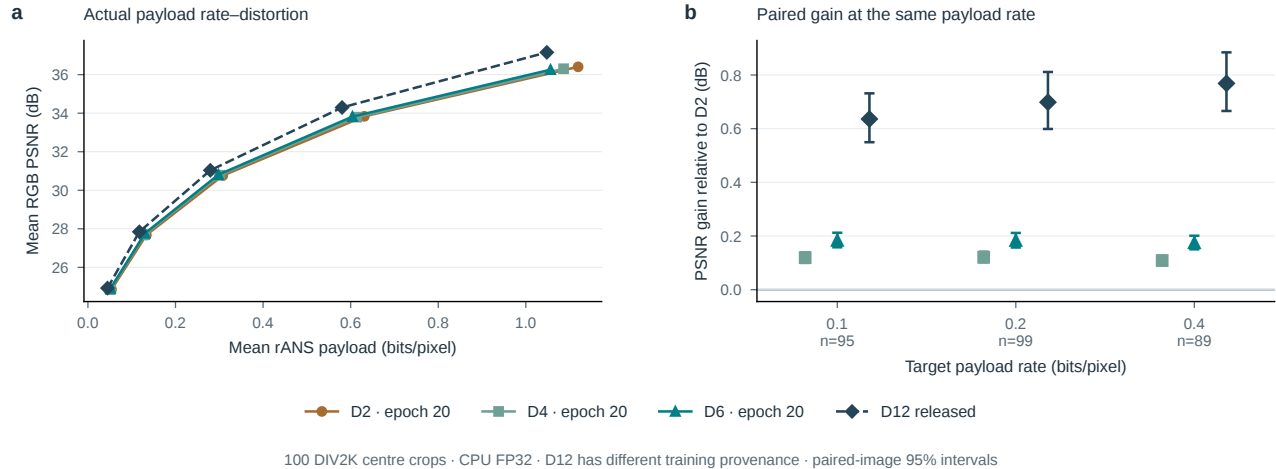


Figure S10. **Shallow models are close, but not identical, at equal coded rate.** **a**, Mean RGB PSNR and actual payload over all 100 crops at five QPs; joining lines are visual guides. **b**, Per-image interpolation and paired differences on four-model common support, with 95% image-bootstrap intervals. D2/D4/D6 are frozen epoch-20 models; released D12 is not matched in training history. These measured streams establish an interim CPU rate–distortion comparison, not final quality or native runtime.

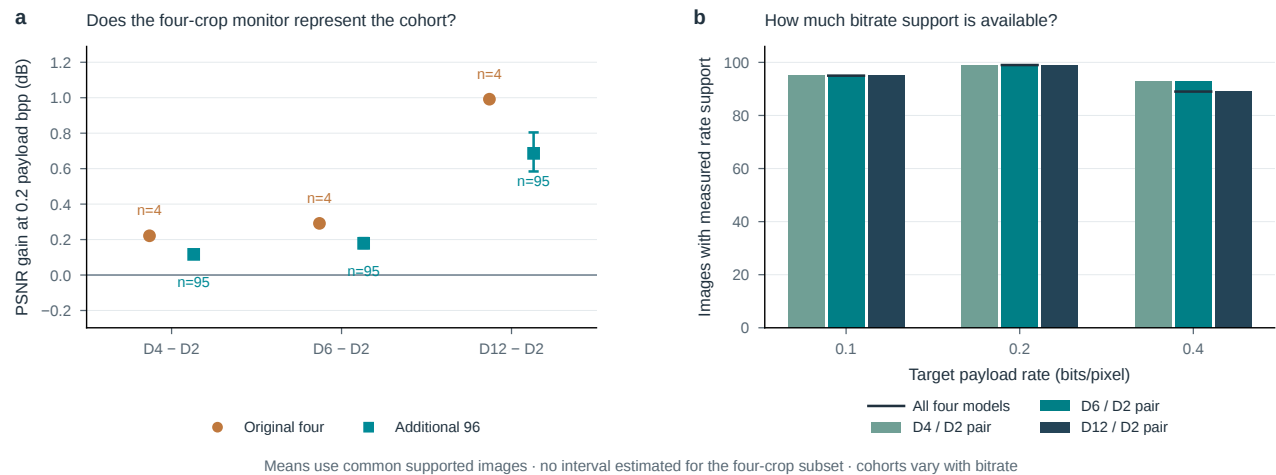


Figure S11. **Coverage is part of the result.** **a**, Original monitor crops and additional images at 0.2 payload bpp, restricted to four-model common support. We do not estimate uncertainty from the four-crop subset. **b**, Pairwise and four-model supported counts at each target rate. A changing cohort is made explicit rather than filled by extrapolation.

516 of all four models' PSNR ranges. All 100 images have a
 517 valid interval. Relative to released D12, mean payload BD-
 518 rate is +21.56% for D2, +17.84% for D4 and +15.69% for
 519 D6 (Fig. S12). This compares unfinished shallow models
 520 with a released checkpoint; it is not the bitrate penalty caused
 521 by truncation at matched convergence. Averaging per-image
 522 BD-rate and comparing population-mean RD curves are dif-
 523 ferent estimands.

524 S12. Does an Additional Expert Add an Addi- 525 tional Choice?

526 A bank with more models has greater capacity, but the value
 527 of its middle experts depends on the attainable allocation
 528 frontier. We perform a conditional leave-D4-out analysis on
 529 the measured epoch-20 D2/D4/D6 curves. The decision unit
 530 is a whole 512^2 crop, not a patch within an image. All three
 531 choices are compared at the same interpolated payload rate
 532 for each image. Dynamic programming minimises summed
 533 RGB MSE under an aggregate neural-decoder Conv2d MAC
 534 budget, permitting unused budget. The objective is reported

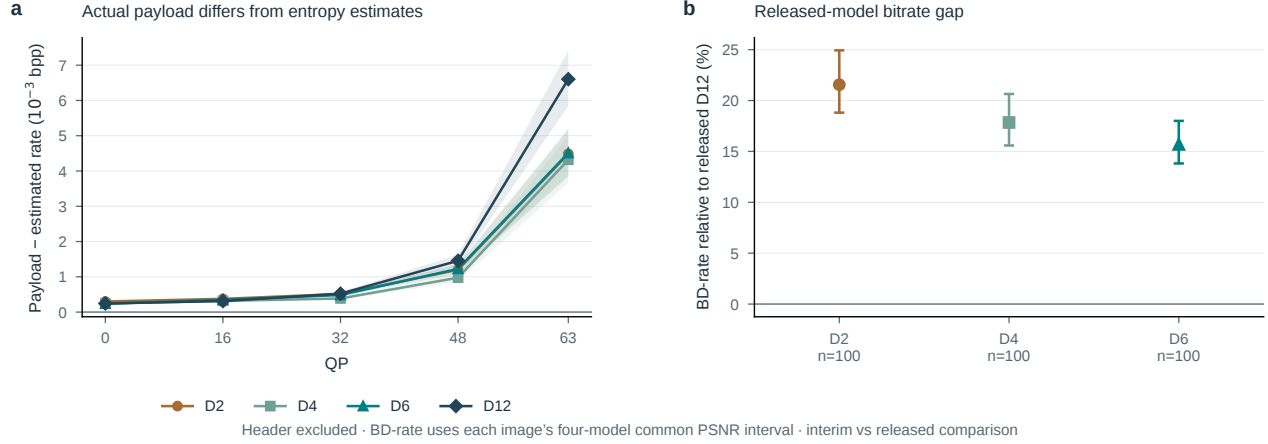


Figure S12. **Estimated entropy and emitted bytes are separate quantities.** **a**, Actual rANS payload minus entropy estimate, excluding the research header. **b**, Mean per-image BD-rate against the released anchor on each image's four-model common PSNR interval. Intervals resample images; no uncertainty from independent training seeds is represented.

as PSNR of pooled MSE, not mean per-image PSNR.

Figure S13 separates two questions. Removing D4 measures its incremental option value relative to the best D2/D6 assignment. Permuting the labels of the best source-calibrated blind histogram measures the value of knowing where to place depth. The blind comparator minimises expected MSE on this same cohort; it is deliberately optimistic and is not a deployable validation-fitted policy. At a budget equal to uniform D4, adding D4 is worth 0.0132, 0.0086 and 0.0061 dB at 0.1, 0.2 and 0.4 bpp. Source-informed placement over the blind expectation is worth 0.0239, 0.0303 and 0.0299 dB. These are deterministic, conditional bounds at one interim checkpoint. They motivate expert-removal controls, rather than establishing that a six-expert spatial bank is necessary.

Complexity is not the training target. We also report every combination of three cheap source statistics, two depth comparisons and three rates: RGB standard deviation, gradient energy and Laplacian energy versus D4–D2 or D6–D2 benefit (Fig. S14). At 0.2 bpp, gradient energy has Spearman correlation 0.270 with D4–D2 but only 0.111 with D6–D2; Laplacian energy has correlation 0.011 with D6–D2. This does not rule out a learned cheap representation. It does caution against treating visible texture as a sufficient label for incremental reconstruction benefit. The statistics are descriptive associations on development-used data, without significance selection or held-out prediction claims. A later selector must be trained and assessed on separate images using rate, distortion and measured cost.

S13. Analytical Scenarios for the Pending Runtime Measurements

The independent depth study has an interim CPU RD diagnostic, but no final trained-model RD or measured native-latency frontier yet. To make the design hypothesis inspectable, Fig. S15 separates exact architectural parameter counts from a hypothetical runtime model. For retained depth d , let f be the assumed fraction of full-depth time spent in the twelve-block trunk and let h be added routing/grouping overhead normalised by the original full-depth time. Under linear trunk-time scaling,

$$\frac{T(d)}{T(12)} = (1 - f) + f \frac{d}{12} + h. \quad (\text{S4})$$

We sweep $f \in \{0.5, 0.7, 0.9\}$ and $h \in \{0, 0.03\}$. These are explicit sensitivity assumptions, not fractions estimated from the hardware. The reciprocal of Eq. (S4) is the plotted scenario speedup. It shows why removing most trunk blocks need not yield a proportional reduction in complete codec time. The curves deliberately contain no predicted PSNR, bitrate or empirical uncertainty bands.

Exact decoder size is 1,767,168 + 1,038,720d parameters; adding the 27,947,520 non-synthesis parameters gives complete-codec size. This formula matches all three independent training manifests. It describes capacity, not compute: retained tensors, arithmetic and measured device time are separate quantities. The data file labels these scenarios explicitly and records every assumption used to generate them. A separate CPU construction of all six depths verifies these counts and exact equality of their retained initial tensors to the seed-matched D12 initialisation; it performs no training or quality evaluation.

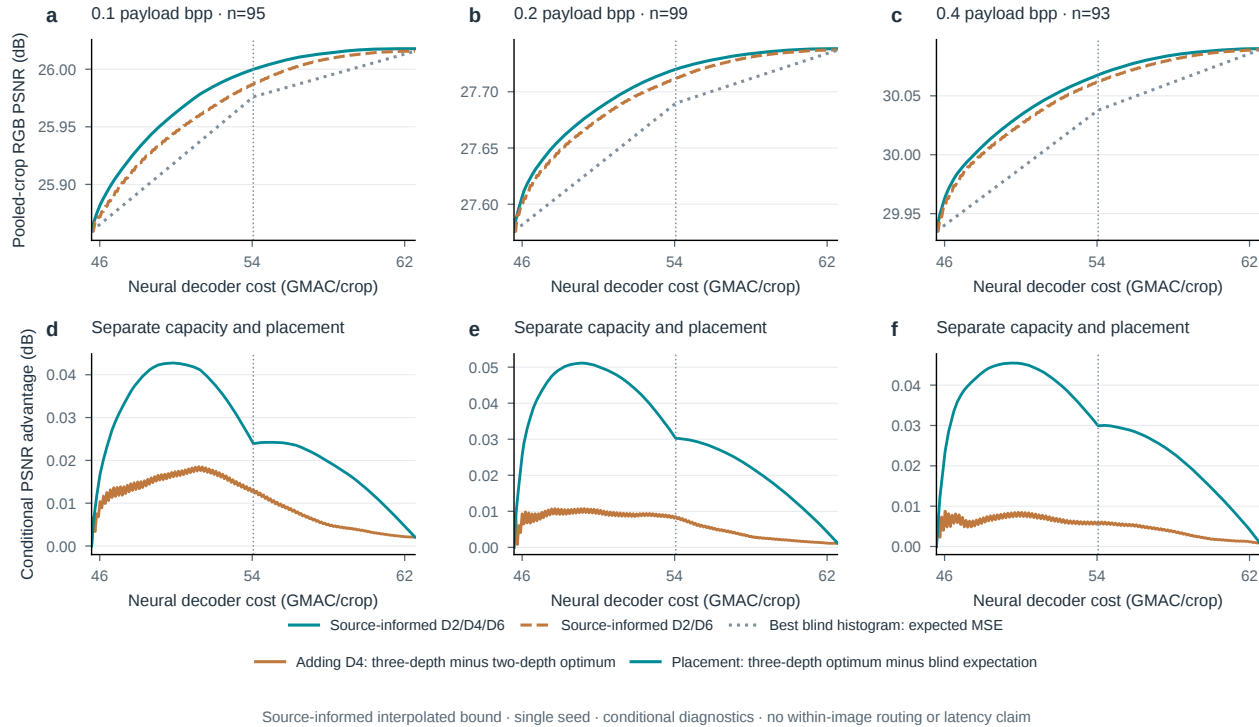


Figure S13. **Separate the value of capacity choices from their assignment.** Top: exact whole-crop allocation under a shared neural-decoder MAC budget. Bottom: incremental D4 option value and assignment premium over the best source-calibrated blind histogram’s expected MSE. Vertical guides mark the uniform-D4 budget. The curves use interpolated equal-rate quality and contain neither patch entropy resets nor a trained MLP. They are not within-image routing results or runtime frontiers.

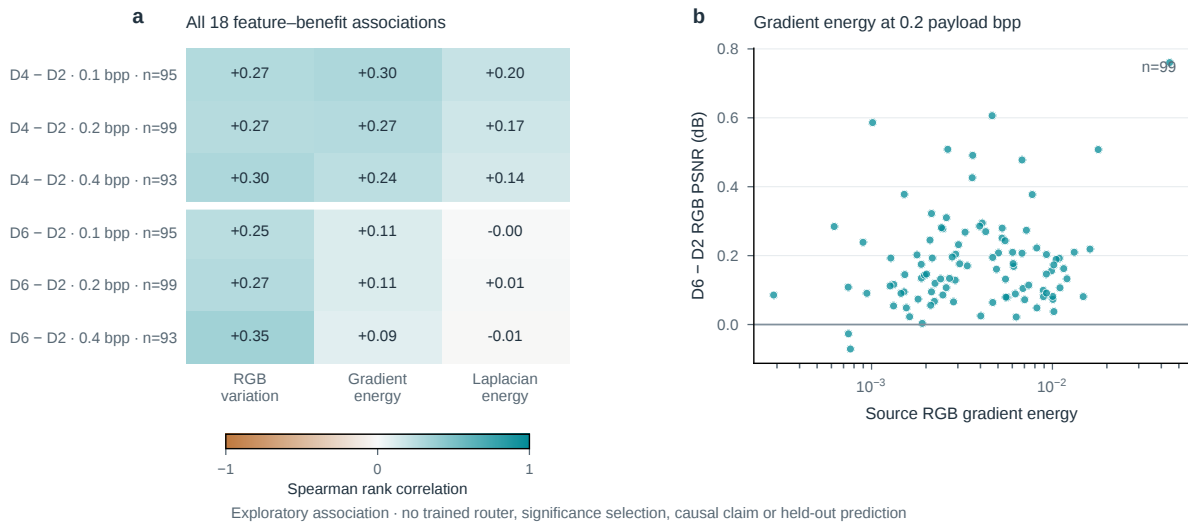


Figure S14. **Cheap complexity measures do not directly identify the useful expert.** All 18 specified feature/depth/rate combinations are shown, with supported image counts. The scatter panel was fixed to gradient energy versus D6-D2 at 0.2 bpp before inspecting the associations. These are correlations with interim equal-rate benefits, not router accuracy or causal feature effects.

S14. Encoder Information Acquisition

Figure S16 separates candidate reconstruction from error-table construction, control search and the final mixed recon-

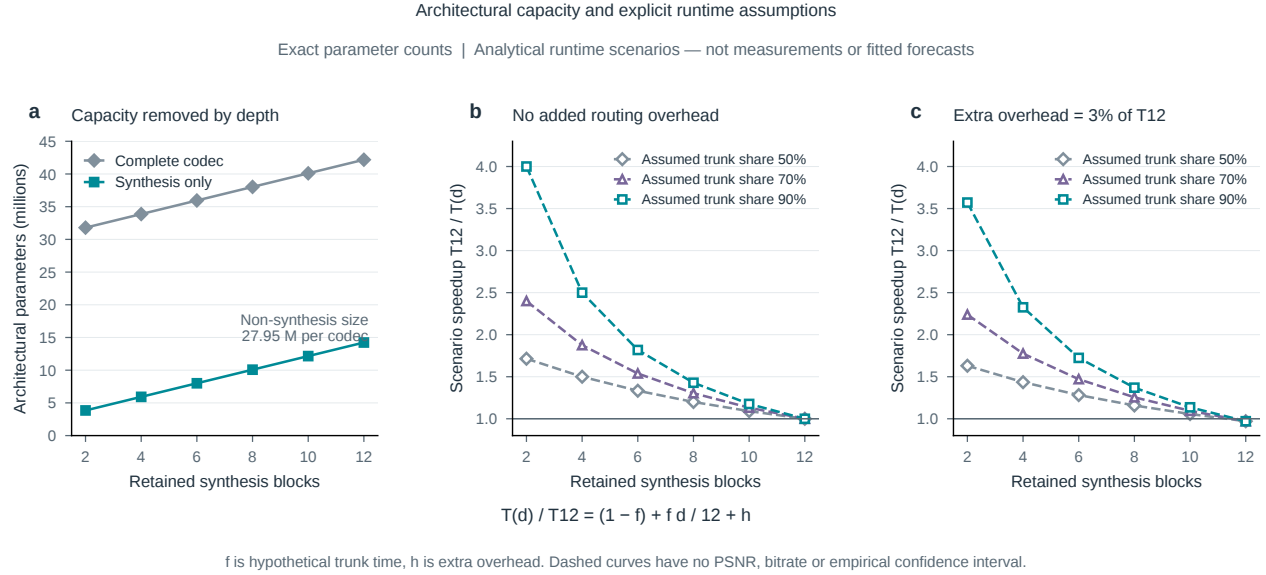


Figure S15. **A transparent design-space illustration for unfinished experiments.** **a**, Exact architectural parameter counts. **b,c**, Analytical runtime sensitivity to assumed trunk-time share and added overhead. Open markers and dashed lines denote assumptions, not measurements or fitted forecasts. No quality or bitrate outcome is assigned to an unfinished model.

struction. A single trunk pass can tap all reachable depths, reducing repeated prefix work. The historical timing summary lacks raw repetition uncertainty; it cannot be used as a paired end-to-end benchmark. In addition, the archived GPU search timer may include pending reference work, so it is excluded from the isolated search comparison. The runtime audit in the paper repository also identifies an older neural replay benchmark whose outputs are not causally reconstructed from decoded stream symbols. Its stage labels must not be read as a verified complete codec.

S15. Fixed Controls on Actual Reconstructions

A stopping policy must eventually be judged on its reconstructed image, rather than only on the table used to choose its control. We therefore replay the sequence-disjoint controls of Sec. S5 at the predeclared QP32 and nominal 0.1 dB target. Every one of the 53 first frames is evaluated with mean and Q90 calibration under router, dither and uniform policies: 318 actual neural reconstructions. The control and exit map are fixed before the source image is opened for reconstruction and scoring. No held-out source-error table or quality-based fallback enters this decision. The checkpoint, sequence folds and development-used corpus remain fixed; this is not an independent test of trained-router generalisation.

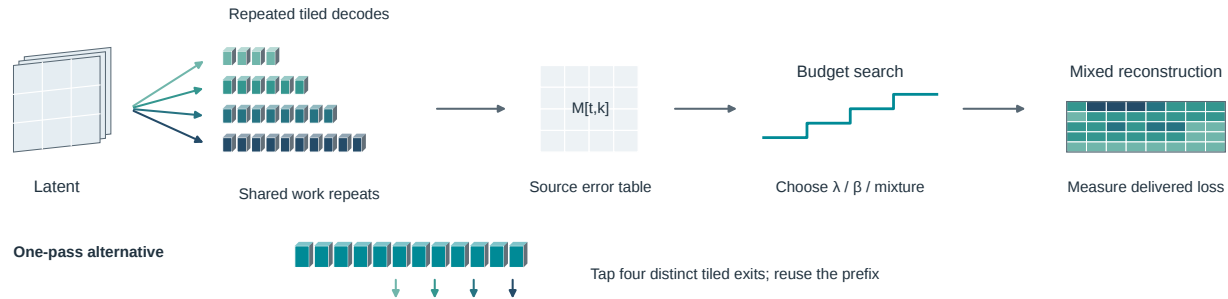
The final reference is the same e15 full-frame reconstruction for every policy. We report cropped, unclipped equal-channel YCbCr loss and separately convert and clip

RGB before computing its MSE-ratio loss. The RGB source is obtained from the archived YCbCr420 input; it is not the original camera RGB. Against this common reference, mean-calibrated router and dither save 25.675% and 23.802% synthesis MACs, at mean RGB losses of 0.08679 and 0.08814 dB. Q90 calibration changes these values to 16.741%/14.204% and 0.05684/0.05636 dB. The corresponding RGB target-exceedance counts are 17/16 under mean control and 4/2 under Q90, each out of 53 images (Fig. S17). A mean or quantile calibration target is therefore not a per-image guarantee.

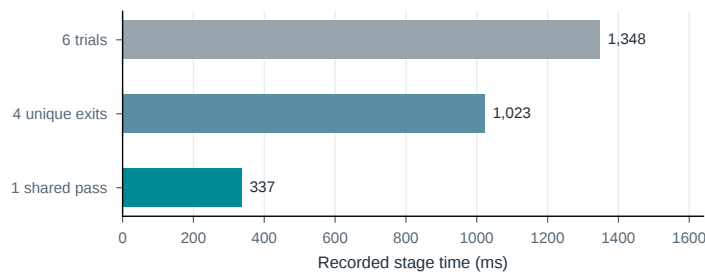
A positive average is not a resolved routing advantage. The paired router-minus-dither saving is 1.87 percentage points under mean control, with a 95% sequence-bootstrap interval of $[-1.60, 5.19]$. Under Q90 it is 2.54 points, with interval $[-0.42, 5.54]$. Both include zero. Moreover, equal control targets do not enforce equal achieved RGB quality. We retain all sequences and report these intervals; this replay does not establish a positive equal-quality routing premium. It complements, rather than replaces, the per-source retrospective delivered-cap comparison in the main paper.

On identical padded support and against the archived scalar reference, the largest absolute discrepancy between actual replay and the stored table prediction is 3.96×10^{-4} dB across all 318 cases. This is a numerical consistency observation. It does not separate backend differences from mixed-head interactions, nor validate entropy-stream decod-

a The source is available at the encoder, but the error table must be computed



b Recorded table construction time



Historical timing: 8 frames $\times$ 3 repetitions at QP32. No new end-to-end latency measurement.

c Calibration changes the cost

Fixed, held-out β

Router uses decoded features.
No per-frame source table is required.

Per-frame source-calibrated β

The plotted budget sweep uses $M[t,k]$.
A fast router does not remove that cost.

Figure S16. **The decision path has an encoder cost.** **a**, Stages required for source-error-based selection, with prefix reuse across candidate exits. **b**, Historical table-construction upper medians over 24 pooled trials, without invented uncertainty bars. **c**, A fixed held-out control and per-frame source calibration require different information. This is a stage analysis, not an end-to-end speedup.

ability. The experiment reconstructs neural outputs on CPU; it is neither a native GPU bitstream test nor a latency measurement.

Recompute the predictor, not only its saved decisions.

A second CPU audit reloads the frozen codec and router checkpoints and recomputes source analysis, shared-stem features and router logits on all 53 frames. Historical GPU and current CPU log-probabilities are not exactly equal: their largest absolute difference is 0.02266. Nevertheless, both fixed cross-fit controls reproduce every exit decision: zero changed tiles out of 1,765 under either mean or Q90 control. This validates the replayed maps at these particular settings, not every control value, QP or backend. Forward hooks count 289.715 router Conv2d/Linear MACs per padded image pixel. Pooling, normalisation, nonlinearities, selection, memory movement and dispatch are excluded; this arithmetic count cannot establish router latency.

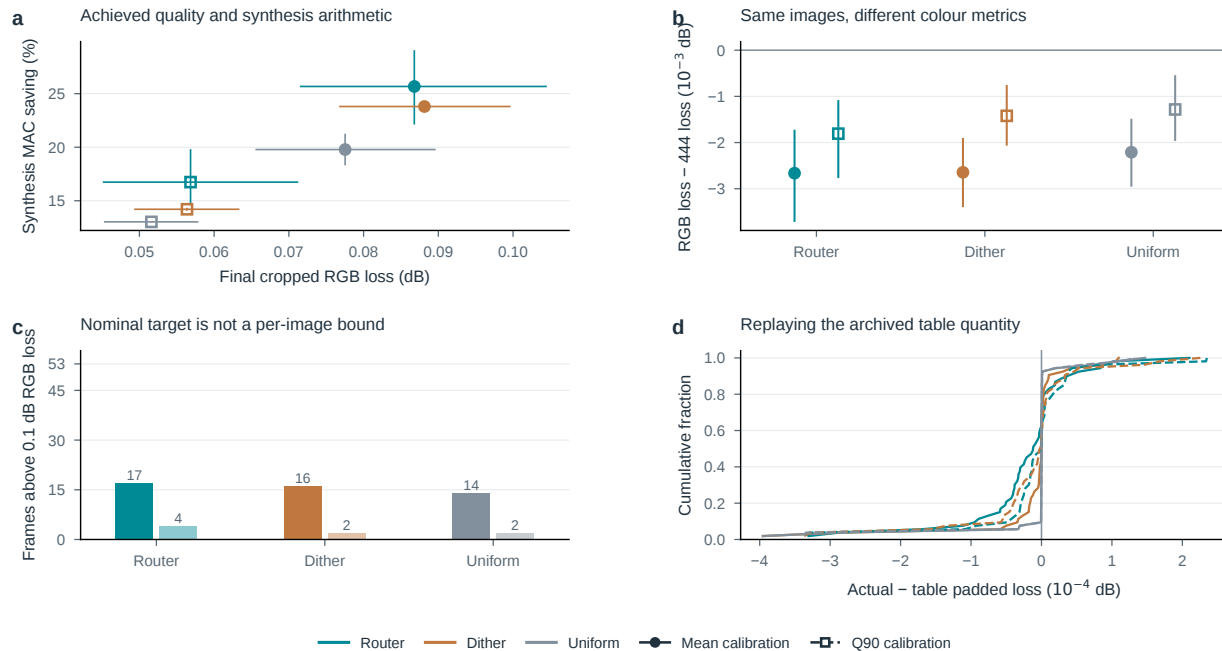
S16. The Cost of Independent Spatial Regions

Shared early exit preserves one encoded representation. A bank of complete codecs changes that constraint: independently encoded regions can repeat source context, reset entropy state and introduce additional headers. Before attributing a gain to expert selection, we measure this cost with the model held fixed.

We predeclare 16 evenly spaced DIV2K centre crops of size 512×512 , five QPs, and three models: D2/D6 at epoch 20 and the released D12 anchor. Each full-frame reconstruction is compared with four independent 256×256 cores, using source-context halos of 0, 32 or 64 pixels. These 240 model-image-QP cases retain every variant. A separate decoder process receives each research bitstream without source pixels; exact reconstruction, latent, quantised-symbol and entropy-index agreement is checked. The CPU FP32 FUFREF1 protocol pads images to multiples of 64. It is not Microsoft's native FP16 CUDA wire format.

Boundary repair can spend more bits than it recovers.

At 0.2 actual payload bpp, all 16 images are supported by all



Matched controls do not imply matched achieved quality · cropped losses use the same e15 dense anchor
CPU neural replay · source RGB derived from YCbCr420 · no bitstream, native latency or external-test claim

Figure S17. Carry the fixed decision through to the actual image. All 53 sequences at QP32, with no outcome-based exclusions. **a**, Achieved RGB loss and synthesis MAC saving with conditional 95% sequence-bootstrap intervals; these points are not an equal-quality frontier. **b**, RGB-minus-444 loss on the same support and reference. **c**, Counts above the nominal numerical threshold; pale bars denote Q90 calibration. **d**, Actual-minus-table loss on padded support. Mean and Q90 controls use a different calibration reference and metric from the cropped RGB report. The paired router–dither saving intervals include zero under both controls.

four D6 curves. Relative to the same model on the full frame, independent cores lose 0.0611 dB RGB PSNR; 32- and 64-pixel halos lose 1.2914 and 1.6666 dB (Fig. S18). These comparisons interpolate each image in log-rate space on common measured support, without extrapolation. The respective 95% paired-image intervals are $[-0.0725, -0.0497]$, $[-1.4810, -1.1071]$ and $[-1.9113, -1.4178]$ dB. They condition on the small fixed cohort and checkpoint, not training seeds.

The same-QP diagnostic explains why a visually improved boundary need not improve the rate–distortion comparison. At D6/QP32, halo32 changes mean PSNR by only +0.0020 dB, while increasing payload by 39.72% on average; halo64 changes it by -0.0037 dB and increases payload by 57.62%. The boundary-specific excess MSE falls towards zero, but the context is still coded and transmitted (Fig. S19). This is a joint effect of region support, entropy resets and redundant context, not an isolated causal estimate for one component.

For halo32, each 288-square window is padded to 320; halo64 uses a 320-square window directly. Both therefore process 1.5625 times the full-frame area, although their actual source support and coded payload differ. Four FUFREF1

headers occupy 352 bytes versus 88 bytes for the full frame, an additional 0.00806 bpp at this crop size. At 0.2 complete-container bpp, the D6 core-only loss is 0.2230 dB on the 15-image common support. The research identity headers are intentionally explicit and are not a minimum production signalling format.

The practical implication is to evaluate how regions are executed alongside which expert is assigned. A shallow bank cannot recover its expected cost advantage merely by adding context everywhere. Padding location and coalescing of adjacent same-depth regions are separate, predeclared controls; they must preserve the selected pixel depths and account for all coded bytes.

S16.1. Padding location changes the rate penalty

The first control rounds a 288-square context window to 320 pixels before analysis. We repeat all 240 cases with a different geometry: keep the image at 288 pixels and pad its 18-square latent to 20 before hyperanalysis. This matches the native path’s spatial rounding convention in the CPU reference implementation. It is not a native CUDA or FP16 equivalence test. The aligned full-frame, core-only and halo64 cases remain controls; the full-frame payload

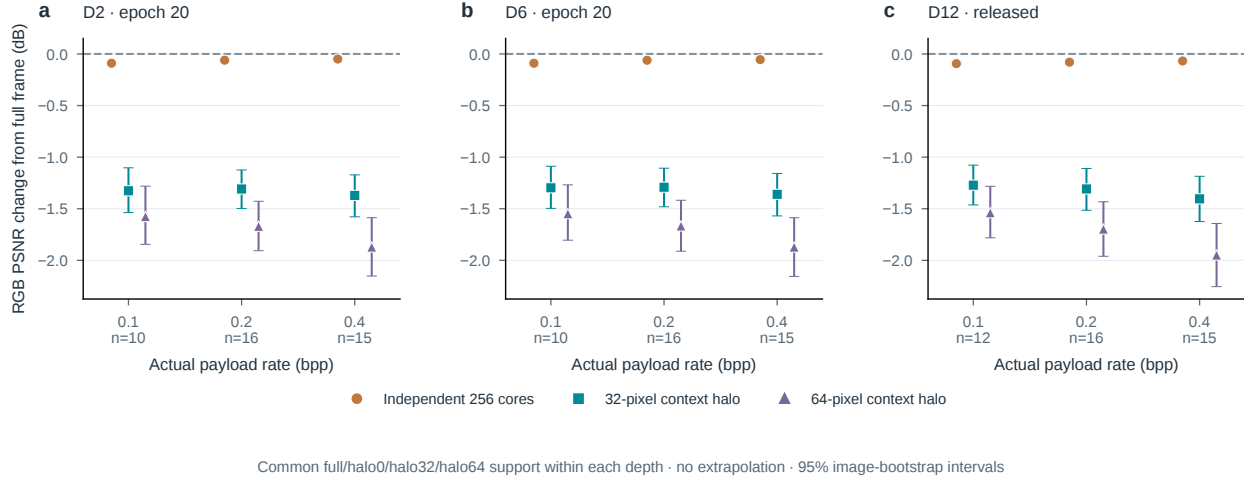


Figure S18. **Charge repeated context to the transmitted rate.** RGB PSNR changes from the same full-frame model at matched actual payload rate. Each depth uses the common supported images across full frame and all three patch variants; counts are shown and the vertical scale is shared. Intervals resample paired images. D2/D6 are interim epoch 20 models; D12 has different released training provenance. The experiment holds depth fixed within every contrast and makes no adaptive-routing or runtime claim.

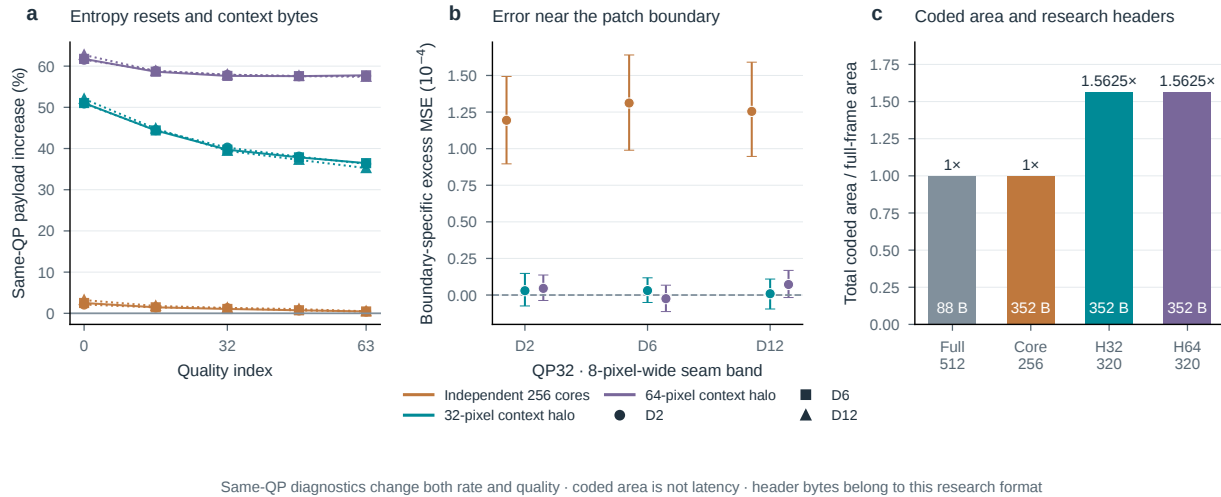
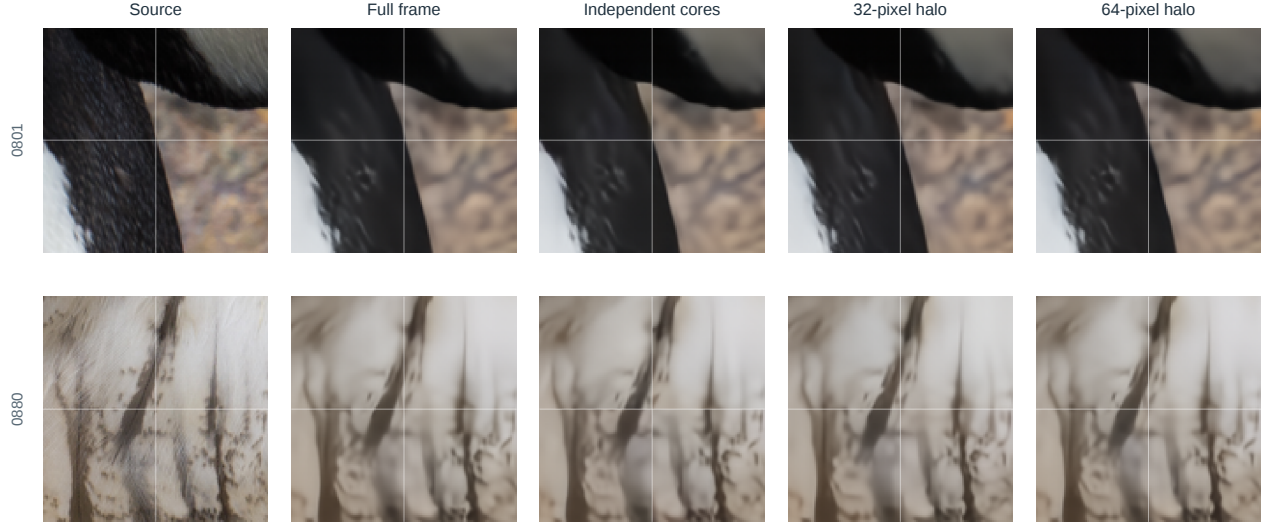


Figure S19. **A better seam and a better codec are different outcomes.** **a**, Same-QP payload increase over the full-frame model. **b**, At QP32, patch-minus-full MSE in the central 8-pixel seam band, minus the corresponding interior excess. **c**, Exact coded area and research-header bytes. Halo32 and halo64 have equal coded area under this padding policy, but different real context. Area ratios are not latency ratios, and same-QP seam diagnostics are not equal-rate quality comparisons.

and reconstruction are verified unchanged.

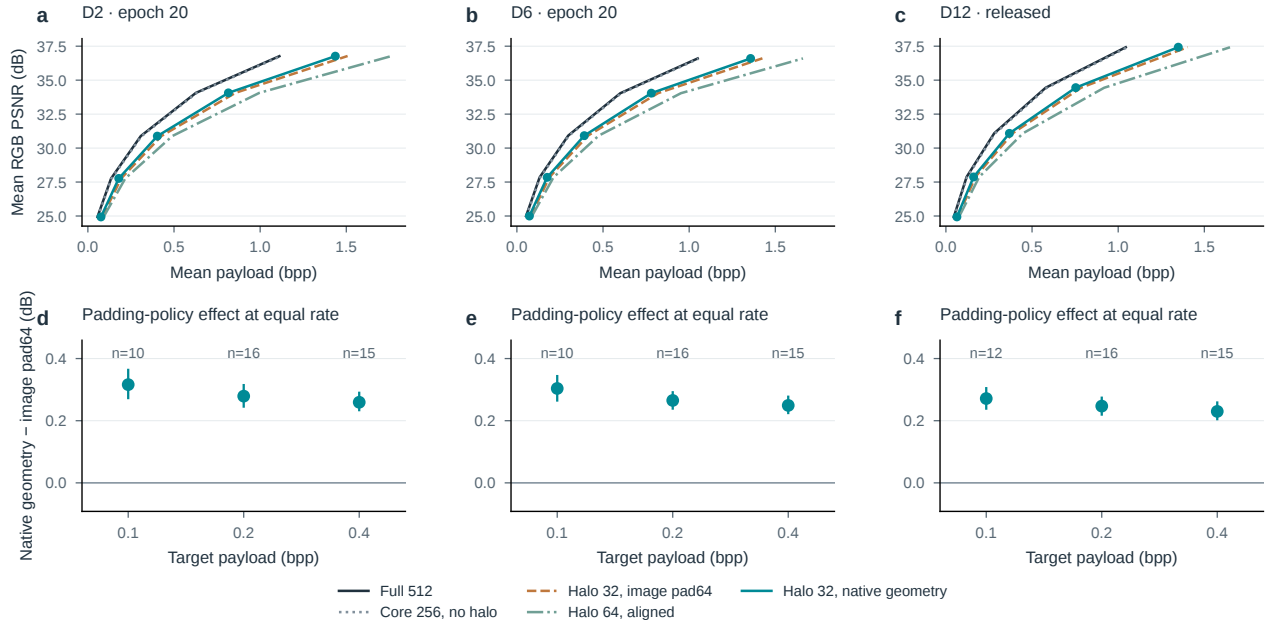
At 0.2 payload bpp, all 16 images support all five variants. Changing the padding location improves halo32 RGB PSNR by 0.2791, 0.2652 and 0.2470 dB for D2, D6 and released D12, respectively (Fig. S21). For D6, the paired 95% interval is $[0.2354, 0.2951]$ dB. This recovery does not remove the cost of independently coded context: the corrected halo32 path still trails the same full-frame D6 by 1.0262 dB, with interval $[0.8614, 1.1915]$. At QP32, the geometry change

reduces payload by 6.03% while changing mean PSNR by -0.0055 dB. These are distinct comparisons: same-QP byte reduction and matched-rate quality improvement. Padding location changes analysis support and hyperprior context together, so the experiment cannot attribute the effect to either mechanism alone.



Images and view window fixed before outcomes · 8-bit display only · light cross marks the patch boundary

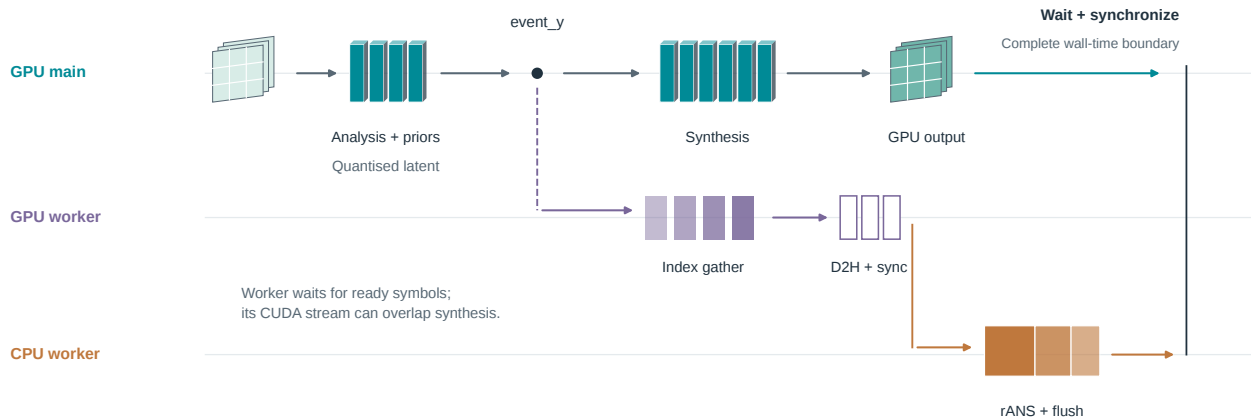
Figure S20. **Predeclared views at the region intersection.** D6 epoch 20, QP32, DIV2K 0801 and 0880. All panels use rows and columns 192:320 of the same 512-square crop; images and window were fixed before outcomes. The common light cross marks the patch boundary. Display images are rounded to 8 bit; numerical metrics use floating-point RGB. These are same-QP views, not equal-bitrate reconstructions.



CPU FP32 geometry control only · native-shaped is not native CUDA output, format or latency

Figure S21. **Keep spatial rounding inside the codec contract.** a–c, Actual-payload RD points averaged over the fixed 16-image cohort, with shared axis limits. d–f, Per-image matched-rate gain from moving padding to the latent, with paired 95% image-bootstrap intervals. Each rate uses the common support of all five variants within that depth; n denotes supported images out of 16, with no extrapolation. The latent-padding implementation uses CPU FP32 research streams; “native geometry” does not denote native CUDA output or latency.

a One API call contains overlapping work



Dependency order only - positions and segment lengths do not represent elapsed time

Figure S22. **Time the completed workload across streams.** Source-derived dependencies of the pinned native DCVC-UF intra encoder. The main-stream reconstruction can overlap worker-stream symbol handling and CPU entropy coding after the latent-ready event. The right boundary includes both the worker wait and completion of the GPU reconstruction. Positions, segment lengths and glyph counts are illustrative, not measured durations or layer counts. This diagram establishes a timing contract, not a speedup.

S16.2. The encoder has a critical path, not a sum of stage costs

The native intra codec also constrains how an adaptive bank should be timed. At the pinned Microsoft revision, analysis and prior processing produce the quantized latent on the main CUDA stream. An event then releases an entropy worker, while synthesis continues on the main stream. The worker gathers symbols, transfers them to the host and performs CPU rANS encoding. The compression call waits for that worker and returns the reconstruction as a GPU tensor; completing the GPU work requires caller-side synchronization. Figure S22 records these source-level dependencies. They permit overlap but do not establish how much overlap occurs on a device. Consequently, a sum of isolated stage times is not the encoder's elapsed time.

For a resident expert bank, the measurement boundary must additionally include source-feature extraction, prediction, grouping, expert dispatch and region assembly. Transfers belong inside that boundary when required by the chosen output contract. Independently trained experts need their own compatible latents; choosing a depth does not make their representations interchangeable. The prepared native correctness harness checks stock versus patched D12 and fresh-process decoding before any GPU latency claim. Its CPU checkpoint and CDF preflight has passed; its CUDA execution remains pending.

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